

Question:

A 5-year-old boy is brought to the physician by his mother because he does not “listen to her” anymore. The mother also reports that her son cannot concentrate on any tasks lasting longer than just a few minutes. Teachers at his preschool report that the patient is more active compared to other preschoolers, frequently interrupts or bothers other children, and is very forgetful. Last year the patient was expelled from another preschool for hitting his teacher and his classmates when he did not get what he wanted and for being disruptive during classes. He was born at term via vaginal delivery and has been healthy except for 3 episodes of acute otitis media at the age of 2 years. He has met all developmental milestones. His mother has major depressive disorder and his father has Graves' disease. He appears healthy and well nourished. Examination shows that the patient does not seem to listen when spoken to directly. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate next step in treatment?

Answer Choices:

- A. Behavior therapy
- B. Methimazole
- C. Fluoxetine
- D. Hearing aids

Reasoning + Answer:

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**Finding reasoning paths : **

- 1 . Evaluate the symptoms (in ability to concentrate , impulsivity , hyper activity , forgetfulness , etc .)
- 2 . Consider the developmental and behavioral context of the patient .
- 3 . Explore psychiatric or neurological conditions that might explain the symptoms .
- 4 . Assess the family history for any relevant genetic or environmental factors .
- 5 . Determine the most appropriate treatment based on the diagnosis .

**Reasoning Process : **

1 . **Symptoms Analysis : **

- The patient exhibits symptoms of hyper activity (in ability to concentrate , impulsivity , restlessness), impulsivity (interrupting and bothering others), and forgetfulness . These symptoms are consistent with Attention Deficit Hyper activity Disorder (ADHD). The history of being expelled from preschool and behaving disruptively further supports this diagnosis .

2 . **Developmental and Behavioral Context : **

- The patient's behavior and symptoms are observable at school and home , indicating that they are not due to environmental or situational factors alone . ADHD is a common condition in children and has a significant impact on behavior and academic performance .

3 . **Psychiatric and Neurological Considerations : **

- ADHD is a neurological condition that affects the prefrontal cortex , leading to difficulties with impulse control , attention , and memory . There is no evidence in the patient's history or presentation to suggest a neurological or psychiatric disorder other than ADHD .

4 . **Family History : **

- The patient's mother has major depressive disorder , and his father has Graves ' disease . While these conditions can influence behavior , there is no direct link to ADHD based on the information provided .

5 . **Treatment Options : **

- The most appropriate treatment for ADHD is a combination of behavioral therapy and medication . Behavioral therapy helps the child develop coping strategies and improve self-regulation . Stimulant medications , such as methyl phen idate or am phetamine , are often used to reduce hyper activity and improve attention . However , the decision to use medication should be based on the severity of symptoms and the child's response to non -pharmac ological interventions .

**Conclusion : **

Based on the symptoms and developmental history , the most likely diagnosis is Attention Deficit Hyper activity Disorder . The most appropriate next step in treatment would likely involve a combination of behavioral therapy and possibly pharmac ological intervention with a stimulant medication , such as methyl phen idate , to address the hyper activity and attention issues .

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<answer>
Methimazole
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